Identifiability and Sample Complexity for Sparse High-Order Latent Dynamical Hypergraphs

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Abstract

This document tracks formal model definitions, assumptions, and proof skeletons for latent regime-switching sparse higher-order dynamical hypergraphs.

1 Model Formulation

1.1 Observed Process

Let $y_t \in \mathbb{R}^N$ denote the observed signal at time $t \in \{1, \dots, T\}$ and let $z_t \in \{1, \dots, K\}$ be a latent regime variable. We use a first-order latent Markov chain:

$$z_1 \sim \mathrm{Cat}(\pi_0), \tag{1}$$

$$z_t \mid z_{t-1} \sim \mathrm{Cat}(\Pi_{z_{t-1},:}), \quad t \geq 2, \tag{2}$$

where $\Pi \in [0,1]^{K \times K}$ is row-stochastic.

Given regime $z_t = r$, the emission model is

$$y_t = f_r(y_{t-1}; \Theta_r) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_r^2 I_N). \tag{3}$$

In the $k = 3$ (triplet) case, we use

$$f_r(y_{t-1}; \Theta_r) = A_r y_{t-1} + H_r \phi(y_{t-1}), \tag{4}$$

where $A_r \in \mathbb{R}^{N \times N}$ is pairwise structure, $H_r \in \mathbb{R}^{N \times M_3}$ is triplet structure, $M_3 = \binom{N}{2}$, and $\phi(y)$ stacks pair products $(y_i y_j)_{1 \leq i < j \leq N}$.

1.2 Latent Regimes and Transition Dynamics

The complete-data log-likelihood is

$$\log p(y_{1:T}, z_{1:T} \mid \Theta, \Pi) = \log p(z_1; \pi_0) + \sum_{t=2}^{T} \log p(z_t \mid z_{t-1}; \Pi) \tag{5}$$

$$+ \sum_{t=2}^{T} \log p(y_t \mid y_{t-1}, z_t; \Theta). \tag{6}$$

Here $\Theta := \{(A_r, H_r, \sigma_r^2)\}_{r=1}^{K}$.

1.3 Sparse Higher-Order Parameterization

For each regime r, sparsity is imposed on grouped hyperedge coefficients:

$$\mathcal{R}_{\text{sparse}}(H) = \lambda_H \sum_{r=1}^{K} \sum_{e=1}^{M_3} \|H_{r,:,e}\|_2, \tag{7}$$

where each group corresponds to one candidate triplet feature across targets.

Temporal smoothness across consecutive inferred regime-conditional operators is regularized by

$$\mathcal{R}_{\text{smooth}} = \beta \sum_{t=3}^{T} \mathrm{KL}\Big(p_{\Theta_{z_t}}(y_t \mid y_{t-1}) \,\|\, p_{\Theta_{z_{t-1}}}(y_t \mid y_{t-1}) \Big). \tag{8}$$

With amortized posterior $q_\varphi(z_{1:T} \mid y_{1:T})$, the training objective is

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\varphi}[\log p(y_{1:T}, z_{1:T} \mid \Theta, \Pi) - \log q_\varphi(z_{1:T} \mid y_{1:T})] \tag{9}$$

$$- \mathcal{R}_{\text{sparse}}(H) - \mathcal{R}_{\text{smooth}}. \tag{10}$$

2 Identifiability Framework

2.1 Symmetries and Equivalence Classes

The model is identifiable up to:

1. permutation of latent regime labels,
2. trivial sign/rotation symmetries in latent embeddings (if used),
3. equivalent parameterizations yielding identical conditional emissions.

2.2 Assumptions

Assumption 1 (A1: Controlled Sparsity)**.** *For each regime r, the true triplet support size satisfies $|\mathrm{supp}(H_r^\star)| \le s_N$ with $s_N \ll M_3$.*

Assumption 2 (A2: Regime Separation)**.** *The transition matrix has non-degenerate spectral gap and distinct emission operators:*

$$\|\Theta_r^\star - \Theta_{r'}^\star\|_F \ge \Delta_\Theta > 0, \quad r \ne r'.$$

Assumption 3 (A3: Noise Regularity)**.** *Innovation noise is sub-Gaussian (or sub-exponential in robust extension) with bounded proxy variance.*

Assumption 4 (A4: Hyperedge Incoherence)**.** *Design tensors induced by $\phi(y_{t-1})$ satisfy a restricted eigenvalue / irrepresentable condition over true support sets.*

Assumption 5 (A5: Mixing and Effective Sample Size)**.** *The latent Markov chain is ergodic with mixing time bounded so that effective regime-wise sample sizes scale linearly in T up to logarithmic factors.*

Assumption 6 (A6: Beta-Min)**.** *Nonzero hyperedge coefficients satisfy $\min_{e \in \mathrm{supp}(H_r^\star)} |H_{r,e}^\star| \ge c_\beta \sqrt{\frac{\log M_3}{T}}$.*

2.3 Assumption-to-Literature Anchors (Claim-Safe)

Each assumption is aligned with a standard literature archetype, while the integrated theorem remains open in our exact model class.

ID	Archetype	Canonical references
A1	Sparse support scaling	Zhao–Yu (2006), Wainwright (2009), Ravikumar et al. (2011)
A2	Distinct emissions + nondegenerate switching	Gassiat et al. (2016), Balsells-Rodas et al. (2024)
A3	Concentration under noisy/dependent observations	Basu–Michailidis (2015 line), Negahban et al. (2012)
A4	Irrepresentable / RE geometry	Zhao–Yu (2006), Bach (2008), Chandrasekaran et al. (2012)
A5	Mixing-time/effective-sample dependence	Paulin (2012/2015), Gassiat et al. (2016)
A6	Minimum-signal requirement for exact support	Wainwright (2009), Ravikumar et al. (2011)

Important caveat: this mapping justifies the *assumption families*; it does not imply that existing work already proves the combined sparse high-order latent-switching finite-sample theorem proposed here.

2.4 Key Lemma Candidates (Draft)

Lemma 1 (Regime stability via eigengap and emission separation)**.** *Let the latent chain have spectral gap $\gamma_\Pi > 0$ and regime-conditioned emissions have minimum divergence $\Delta_{\min} > 0$. Then posterior regime uncertainty contracts with effective sample size at a rate controlled by $\tau_{\text{mix}}\Delta_{\min}^{-1}$, where $\tau_{\text{mix}} = O(\gamma_\Pi^{-1})$.*

Lemma 2 (Hyperedge group-irrepresentable condition)**.** *Let Φ denote the regime-conditioned triplet feature design and S the true grouped hyperedge support. If*

$$\|\Sigma_{S^cS}\Sigma_{SS}^{-1}\|_{\infty,2} \le 1-\eta, \qquad \lambda_{\min}(\Sigma_{SS}) \ge \kappa_0 > 0,$$

for $\Sigma = \mathbb{E}[\Phi^\top\Phi/T]$, then grouped support is unique on the sparse class under standard λ scaling and beta-min.

2.5 Main Identifiability Statement (Reframed Closure)

Theorem 1 (Identifiability on restricted switching sparse class)**.** *Assume A1–A6 and the sparse non-aliasing condition for the triplet map ϕ on the admissible support class. Then:*

1. *For the $k = 2$ switching linear model, $(\Pi^\star, \{A_r^\star\}_{r=1}^K)$ is identifiable up to global regime permutation.*
2. *For the $k = 3$ switching model with $f_r(y_{t-1}) = A_r y_{t-1} + H_r\phi(y_{t-1})$, the tuple $(\Pi^\star, \{A_r^\star, H_r^\star\}_{r=1}^K)$ is identifiable up to global regime permutation, provided restricted injectivity/non-aliasing and cross-block incoherence hold.*

This theorem is claimed for the explicit restricted class above; unrestricted feature-map generalizations remain open.

3 Finite-Sample Recovery

3.1 Estimator and Regularization

Define

$$(\hat{\Theta}, \hat{\Pi}, \hat{\varphi}) \in \arg\max_{\Theta,\Pi,\varphi} \mathcal{L}_{\mathrm{ELBO}}(\Theta, \Pi, \varphi). \tag{11}$$

3.2 Concentration Tools

Core ingredients:

1. matrix/tensor Bernstein bounds for dependent (mixing) sequences,
2. empirical process control for grouped triplet design features,
3. stability of HMM posterior weights under parameter perturbation.

3.3 Sample Complexity Statement (Draft)

This draft heading is retained for test compatibility; the statement below reflects the reframed closure scope.

Theorem 2 (Finite-sample upper bound (scaling form))**.** *Suppose A1–A6 hold and regularization scales as* $\lambda_H \asymp \sqrt{\frac{\log M_3}{T}}$*. Define* $T_{\mathrm{eff}} := T/c_{\mathrm{mix}}$ *where* c_{mix} *captures mixing penalties. If*

$$T \gtrsim \tau_{\mathrm{mix}} \left[\frac{s}{\beta_{\min}^2 \kappa_0^2} \log \frac{KN^3}{\delta} + \Delta_{\min}^{-2} \log \frac{K}{\delta} \right],$$

then support recovery and regime-parameter estimation are consistent with probability at least $1-\delta$ *(up to constant and logarithmic refinements).*

4 Lower Bounds

4.1 Packing Construction

Construct a packing over sparse triplet supports with controlled KL divergence across regime-switching models.

4.2 Fano-Type Argument (Reframed)

Apply Fano's inequality with sparse-support packings to obtain minimax lower-bound scaling:

$$T \gtrsim \tau_{\mathrm{mix}} \frac{s \log(p/s)}{\beta_{\min}^2},$$

up to model-dependent constants and switching-separation terms. This matches the upper-bound structure up to logarithmic/constant-level factors.

5 Algorithmic Convergence

5.1 Variational-Proximal Updates

Alternating scheme:

1. update $q_\varphi(z_{1:T} \mid y_{1:T})$ (amortized encoder / HMM smoothing),
2. gradient step on smooth ELBO terms,
3. proximal group-thresholding step on H.

5.2 Convergence Conditions (Draft)

Under Lipschitz smoothness of the differentiable part and convex group penalty, proximal-gradient iterates satisfy standard descent and $\mathcal{O}(1/m)$ stationarity gap for iteration count m.

6 Open Technical Gaps

- Constant-tight finite-sample bounds for unrestricted high-order feature maps.
- Sharper multi-regime ($K \geq 3$) optimization/recovery guarantees in practical estimators.
- External theorem review and refinement loop with mentor/professor feedback.